

Tile It, Split It, Find the Area



Grade 3 • Mathematics
NC.3.MD.7

Connect tiled rectangles with multiplication.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A rectangular display for the science club measures 9 units by 5 units. Draw unit-square rows, find the area, and show a split-and-add check.

$9 \times 5 = 45$. One split gives $(8 \times 5) + (1 \times 5) = 40 + 5 = 45$.



Build a second model together

A rectangular display for the reading group measures 4 units by 8 units. Draw unit-square rows, find the area, and show a split-and-add check.

Discuss: What should the first step find?



Your turn • 1

A rectangular display for the library team measures 7 units by 2 units. Draw unit-square rows, find the area, and show a split-and-add check.

Represent → solve → explain → check



Your turn • 2

A rectangular display for the design team measures 2 units by 2 units. Draw unit-square rows, find the area, and show a split-and-add check.

Represent → solve → explain → check



Your turn • 3

A rectangular display for the community center measures 2 units by 6 units. Draw unit-square rows, find the area, and show a split-and-add check.

Represent → solve → explain → check



Your turn • 4

A rectangular display for the school garden measures 2 units by 2 units. Draw unit-square rows, find the area, and show a split-and-add check.

Represent → solve → explain → check



Exit question • Explain independently

A rectangular display for the design team measures 3 units by 3 units. Draw unit-square rows, find the area, and show a split-and-add check.

Include a check. What would you teach someone else?



Teacher review and next steps

1. 14 square units

Exit: 9 square units

2. 4 square units

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. 12 square units

4. 4 square units

